

## C&I ESS CFE-250H2-PV-STC



### Application Scenarios

Suitable for sites with high power capacity demands, such as large-scale commercial and industrial enterprises, data centers, factories, and mobile energy storage supply.



#### High Power • High Energy Density

Integrated design with 250kW / 522kWh, occupying only 2.7m². System efficiency >88%, supports multi-unit parallel expansion. Meets the energy demands of large-scale commercial and industrial scenarios.



#### PV-Storage Integration • Dual MPPT

Built-in dual 63kW MPPT channels (total power 252kW), weighted efficiency 99%. Enables intelligent energy flow control among PV, storage, load, and grid. Supports flexible AC / DC coupling and retrofit upgrades.



#### Uninterrupted • Highly Reliable Power Supply

Integrated STS (Static Transfer Switch) enables seamless grid-connected / off-grid switching in < 20ms grid-connected / off-grid seamless switching. Ensures continuous power supply for critical loads, with 110% continuous overload capacity and 120% short-term output capability.



#### Intelligent Management • Cloud Collaboration

Two-layer BMS for precise cell management. Built-in BMS with local + cloud system. Enables real-time monitoring, energy dispatch, and remote diagnostics. Compatible with mainstream SCADA platforms.



#### Safety First • Liquid Cooling Protection

Utilizes LFP cells (3.2V 314Ah), cycle life >8000 cycles. Equipped with liquid cooling system, dual airflow isolation, and multiple electrical protections. Comprehensive protection against overcurrent, short circuits, and arc faults.



#### Outdoor-Rated Design • Global Compatibility

Rated IP54 + C4, operational temperature range -20°C ~ +55°C. Factory-integrated, plug-and-play. Complies with IEC international standards, adaptable to global grid environments.

| Model                            | CFE 250H2-PV-STC                     |
|----------------------------------|--------------------------------------|
| Battery Parameter                |                                      |
| Battery Type                     | LFP(3.2 V 314 Ah)                    |
| Battery System Configuration     | 2P260S                               |
| Battery System Capacity          | 522.48 kWh                           |
| Battery Voltage Range            | 728-949 V                            |
| Battery Cycle Life               | 8,000 cycles                         |
| AC On-Grid                       |                                      |
| The Rated Output                 | 250 kW                               |
| Overload Capacity                | 110% continuous; 120% for ≤1 minute. |
| Rated Voltage                    | AC 400 V                             |
| Rated Output Current             | 360 A                                |
| AC Access Mode                   | Three-phase four-wire                |
| Isolation Method                 | Non – isolated                       |
| Grid Voltage Range               | 380 V(-20%~+15%)                     |
| Grid Frequency Range             | 50 Hz / 60 Hz±2.5 Hz                 |
| Total Harmonic Distortion        | ≤3% (Full Load)                      |
| Power Factor                     | -0.99 – 0.99                         |
| DC Component                     | ≤0.5%                                |
| Charge/Discharge Conversion Time | <100 ms                              |
| MPPT Parameters                  |                                      |
| MPPT Rated Power                 | 252 kW                               |
| Number of MPPT Trackers          | Four MPPT trackers, 4 x 63 kW        |
| MPPT Efficiency                  | 99%                                  |
| PV Input Voltage Range           | 394 – 680 V                          |
| Maximum Operating Current        | 360 A                                |
| AC Off-Grid Parameters           |                                      |
| AC Off-grid Voltage              | AC400V                               |
| AC Voltage Range                 | AC400V±3%                            |
| AC Off-grid Frequency            | 50Hz / 60Hz                          |
| Off-grid Output THDU             | ≤3% (linear load)                    |
| Unbalanced Load Capacity         | 100%                                 |
| System Parameter                 |                                      |
| System Efficiency                | >88%                                 |
| Size (W × D × H)                 | 2,365 × 1,975 × 1,373 mm             |
| Weight                           | 5T                                   |
| Operating Temperature            | -20 to 55 °C                         |
| Environment Humidity             | 0–95% RH (non-condensing)            |
| Work Altitude                    | <3000m                               |
| Protection Level                 | IP54                                 |
| Corrosion Protection Level       | C4                                   |
| Cooling Method                   | Liquid cooling                       |
| Fire Suppression System          | Aerosol                              |